Principles of Programming Languages
Assignment 3 – Part 1

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Question 1.1

For each typing statement, we determine whether it is true or false.

1.

$$\{f : (T_1 \to T_2),\ g : (T_1 \to T_2),\ a : T_1\} \vdash (f\ (g\ a)) : T_2$$

Answer: False.
Since

$$g : (T_1 \to T_2) \qquad \text{and} \qquad a : T_1$$

we get:

$$(g\ a) : T_2$$

However,

$$f : (T_1 \to T_2)$$

so f expects an argument of type T_1. In the expression

$$(f\ (g\ a))$$

the argument passed to f has type T_2, not T_1. Since T_1 and T_2 are not necessarily the same type, the expression is not well-typed.
Therefore, the typing statement is false.

2.

$$\{f : (T_1 * T_2 \to T_3)\} \vdash (\text{lambda}\ (x)\ (f\ x\ y)) : (T_2 \to T_3)$$

Answer: False.
The variable y appears free in the body:

$$(f\ x\ y)$$

but y is not assigned any type in the typing environment. Therefore, we cannot type-check the body of the lambda expression.

Also, the claimed type of the lambda expression is:

$$(T_2 \to T_3)$$

so x should have type T_2. However,

$$f : (T_1 * T_2 \to T_3)$$

so the first argument of f should have type T_1. Therefore, the typing statement is false.

3.

$$\{f : (T_1 * T_2 \to T_3),\ y : T_2\} \vdash (\text{lambda } (x)\ (f\ x\ y)) : (T_1 \to T_3)$$

Answer: True.

The environment contains:

$$f : (T_1 * T_2 \to T_3) \qquad \text{and} \qquad y : T_2$$

In the expression

$$(\text{lambda } (x)\ (f\ x\ y))$$

if

$$x : T_1$$

then the application

$$(f\ x\ y)$$

is valid, because f receives arguments of types T_1 and T_2. Therefore,

$$(f\ x\ y) : T_3$$

Thus, the whole lambda expression has type:

$$(T_1 \to T_3)$$

So the typing statement is true.

Question 1.2

We perform type inference manually using the Type Equations method.

Expression 1

$$(\text{lambda } (f_1\ g_1\ x_1\ y_1)\ (f_1\ (g_1\ y_1\ x_1)\ (g_1\ x_1\ y_1)))$$

i. Pool of fresh type variables

$$\begin{array}{lcl}
T0 & : & (\text{lambda } (f_1\ g_1\ x_1\ y_1)\ (f_1\ (g_1\ y_1\ x_1)\ (g_1\ x_1\ y_1))) \\
T1 & : & (f_1\ (g_1\ y_1\ x_1)\ (g_1\ x_1\ y_1)) \\
T2 & : & (g_1\ x_1\ y_1) \\
T3 & : & (g_1\ y_1\ x_1) \\
Tf1 & : & f_1 \\
Tg1 & : & g_1 \\
Tx1 & : & x_1 \\
Ty1 & : & y_1
\end{array}$$

ii. Type equations

From the lambda expression:

$$T0 = (Tf1 * Tg1 * Tx1 * Ty1 \to T1)$$

From the application

$$(f_1\ (g_1\ y_1\ x_1)\ (g_1\ x_1\ y_1))$$

we get:

$$Tf1 = (T3 * T2 \to T1)$$

From the application

$$(g_1\ x_1\ y_1)$$

we get:

$$Tg1 = (Tx1 * Ty1 \to T2)$$

From the application

$$(g_1\ y_1\ x_1)$$

we get:

$$Tg1 = (Ty1 * Tx1 \to T3)$$

Therefore, the full set of equations is:

$$\begin{array}{ll} 1. & T0 = (Tf1 * Tg1 * Tx1 * Ty1 \to T1) \\ 2. & Tf1 = (T3 * T2 \to T1) \\ 3. & Tg1 = (Tx1 * Ty1 \to T2) \\ 4. & Tg1 = (Ty1 * Tx1 \to T3) \end{array}$$

iii. Unification

From equations 3 and 4:

$$(Tx1 * Ty1 \to T2) = (Ty1 * Tx1 \to T3)$$

Comparing the parameter types and return types gives:

$$Tx1 = Ty1 \qquad \text{and} \qquad T2 = T3$$

Now substitute these results into the other equations.
From

$$Tf1 = (T3 * T2 \to T1)$$

and $T3 = T2$, we get:

$$Tf1 = (T2 * T2 \to T1)$$

From

$$Tg1 = (Tx1 * Ty1 \to T2)$$

and $Ty1 = Tx1$, we get:

$$Tg1 = (Tx1 * Tx1 \to T2)$$

Finally, substitute into the type of the whole lambda:

$$T0 = (Tf1 * Tg1 * Tx1 * Ty1 \to T1)$$

Using

$$Tf1 = (T2 * T2 \to T1)$$

$$Tg1 = (Tx1 * Tx1 \to T2)$$

and

$$Ty1 = Tx1$$

we get:

$$T0 = ((T2 * T2 \to T1) * (Tx1 * Tx1 \to T2) * Tx1 * Tx1 \to T1)$$

Final substitution

$$\begin{aligned} Ty1 &= Tx1 \\ T3 &= T2 \\ Tf1 &= (T2 * T2 \to T1) \\ Tg1 &= (Tx1 * Tx1 \to T2) \end{aligned}$$

Final inferred type

$$T0 = ((T2 * T2 \to T1) * (Tx1 * Tx1 \to T2) * Tx1 * Tx1 \to T1)$$

Expression 2

$$((\text{lambda } (f_1) \ (\text{lambda } (x_1) \ (f_1 \ x_1 \ x_1))) \ (\text{lambda } (y_1) \ y_1))$$

i. Pool of fresh type variables

$$\begin{array}{lcl}
T0 & : & ((\text{lambda } (f_1) \ (\text{lambda } (x_1) \ (f_1 \ x_1 \ x_1))) \ (\text{lambda } (y_1) \ y_1)) \\
T1 & : & (\text{lambda } (f_1) \ (\text{lambda } (x_1) \ (f_1 \ x_1 \ x_1))) \\
T2 & : & (\text{lambda } (y_1) \ y_1) \\
T3 & : & (\text{lambda } (x_1) \ (f_1 \ x_1 \ x_1)) \\
T4 & : & (f_1 \ x_1 \ x_1) \\
Tf1 & : & f_1 \\
Tx1 & : & x_1 \\
Ty1 & : & y_1
\end{array}$$

ii. Type equations

From the application of $T1$ to $T2$:

$$T1 = (T2 \to T0)$$

From the lambda expression

$$(\text{lambda } (f_1) \ (\text{lambda } (x_1) \ (f_1 \ x_1 \ x_1)))$$

we get:

$$T1 = (Tf1 \to T3)$$

From the lambda expression

$$(\text{lambda } (y_1) \ y_1)$$

we get:

$$T2 = (Ty1 \to Ty1)$$

From the lambda expression

$$(\text{lambda } (x_1) \ (f_1 \ x_1 \ x_1))$$

we get:

$$T3 = (Tx1 \to T4)$$

From the application

$$(f_1\ x_1\ x_1)$$

we get:

$$Tf1 = (Tx1 * Tx1 \to T4)$$

Therefore, the full set of equations is:

$$\begin{array}{ll} 1. & T1 = (T2 \to T0) \\ 2. & T1 = (Tf1 \to T3) \\ 3. & T2 = (Ty1 \to Ty1) \\ 4. & T3 = (Tx1 \to T4) \\ 5. & Tf1 = (Tx1 * Tx1 \to T4) \end{array}$$

iii. Unification

From equations 1 and 2:

$$(T2 \to T0) = (Tf1 \to T3)$$

we get:

$$T2 = Tf1 \qquad \text{and} \qquad T0 = T3$$

Now, from equation 3:

$$T2 = (Ty1 \to Ty1)$$

and from equation 5:

$$Tf1 = (Tx1 * Tx1 \to T4)$$

Since we also got $T2 = Tf1$, we must unify:

$$(Ty1 \to Ty1) = (Tx1 * Tx1 \to T4)$$

This equation cannot be solved, because the left side is a procedure type with one parameter, while the right side is a procedure type with two parameters.

Final result

The inference fails.

The failure is triggered by the equation:

$$(Ty1 \to Ty1) = (Tx1 * Tx1 \to T4)$$

This equation attempts to unify a procedure with one parameter with a procedure with two parameters. Therefore, the expression is not typable.

Expression 3

$$((\text{lambda }(f_1)\ (\text{lambda }(x_1)\ ((f_1\ x_1)\ x_1)))\ (\text{lambda }(y_1)\ y_1))$$

i. Pool of fresh type variables

$$\begin{aligned}
T0 &: ((\text{lambda }(f_1)\ (\text{lambda }(x_1)\ ((f_1\ x_1)\ x_1)))\ (\text{lambda }(y_1)\ y_1)) \\
T1 &: (\text{lambda }(f_1)\ (\text{lambda }(x_1)\ ((f_1\ x_1)\ x_1))) \\
T2 &: (\text{lambda }(y_1)\ y_1) \\
T3 &: (\text{lambda }(x_1)\ ((f_1\ x_1)\ x_1)) \\
T4 &: ((f_1\ x_1)\ x_1) \\
T5 &: (f_1\ x_1) \\
Tf &: f_1 \\
Tx &: x_1 \\
Ty &: y_1
\end{aligned}$$

ii. Type equations

From the whole application:

$$T1 = (T2 \to T0)$$

From the lambda expression

$$(\text{lambda }(f_1)\ (\text{lambda }(x_1)\ ((f_1\ x_1)\ x_1)))$$

we get:

$$T1 = (Tf \to T3)$$

From the lambda expression

$$(\text{lambda }(y_1)\ y_1)$$

we get:

$$T2 = (Ty \to Ty)$$

From the lambda expression

$$(\text{lambda }(x_1)\ ((f_1\ x_1)\ x_1))$$

we get:

$$T3 = (Tx \to T4)$$

From the application

$$((f_1\ x_1)\ x_1)$$

we get:

$$T5 = (Tx \to T4)$$

From the application

$$(f_1\ x_1)$$

we get:

$$Tf = (Tx \to T5)$$

Therefore, the full set of equations is:

1. $T1 = (T2 \to T0)$
2. $T1 = (Tf \to T3)$
3. $T2 = (Ty \to Ty)$
4. $T3 = (Tx \to T4)$
5. $T5 = (Tx \to T4)$
6. $Tf = (Tx \to T5)$

iii. Unification

From equations 1 and 2:

$$(T2 \to T0) = (Tf \to T3)$$

we get:

$$T2 = Tf \qquad \text{and} \qquad T0 = T3$$

From equation 3:

$$T2 = (Ty \to Ty)$$

and since $T2 = Tf$, we get:

$$Tf = (Ty \to Ty)$$

But from equation 6:

$$Tf = (Tx \to T5)$$

Therefore, we must unify:

$$(Ty \to Ty) = (Tx \to T5)$$

This gives:

$$Ty = Tx \qquad \text{and} \qquad Ty = T5$$

So:

$$Tx = Ty \qquad \text{and} \qquad T5 = Ty$$

Now use equation 5:

$$T5 = (Tx \to T4)$$

Substituting $T5 = Ty$ and $Tx = Ty$, we get:

$$Ty = (Ty \to T4)$$

This equation cannot be solved because Ty occurs inside the type $(Ty \to T4)$. This violates the occurs check and would require an infinite type.

Final result

The inference fails.

The failure is triggered by the equation:

$$Ty = (Ty \to T4)$$

This equation fails the occurs check, because Ty occurs inside the type $(Ty \to T4)$. Therefore, unification fails and the expression is not typable.

Question 1.3

Sum Types

Expression:

$$(\text{lambda }(f)\ (f\ (right\ (\text{lambda }(s)\ (f\ (left\ s))))))$$

i. Pool of fresh type variables

$$\begin{aligned}
T0 &: (\text{lambda }(f)\ (f\ (right\ (\text{lambda }(s)\ (f\ (left\ s)))))) \\
T1 &: (f\ (right\ (\text{lambda }(s)\ (f\ (left\ s))))) \\
T2 &: (right\ (\text{lambda }(s)\ (f\ (left\ s)))) \\
T3 &: (\text{lambda }(s)\ (f\ (left\ s))) \\
T4 &: (f\ (left\ s)) \\
T5 &: (left\ s) \\
Tf &: f \\
Ts &: s
\end{aligned}$$

We also introduce two fresh type variables for the unknown sides of the sum types:

$$Tright \qquad \text{and} \qquad Tleft$$

ii. Type equations

From the outer lambda:

$$T0 = (Tf \rightarrow T1)$$

From the application

$$(f\ (right\ (\text{lambda }(s)\ (f\ (left\ s)))))$$

we get:

$$Tf = (T2 \rightarrow T1)$$

From the expression

$$(right\ (\text{lambda }(s)\ (f\ (left\ s))))$$

using the rule for $right$, we get:

$$T2 = (Tright + T3)$$

From the inner lambda

$$(\text{lambda } (s)\ (f\ (left\ s)))$$

we get:

$$T3 = (Ts \to T4)$$

From the application

$$(f\ (left\ s))$$

we get:

$$Tf = (T5 \to T4)$$

From the expression

$$(left\ s)$$

using the rule for $left$, we get:

$$T5 = (Ts + Tleft)$$

Therefore, the full set of equations is:

$$\begin{array}{ll} 1. & T0 = (Tf \to T1) \\ 2. & Tf = (T2 \to T1) \\ 3. & T2 = (Tright + T3) \\ 4. & T3 = (Ts \to T4) \\ 5. & Tf = (T5 \to T4) \\ 6. & T5 = (Ts + Tleft) \end{array}$$

iii. Unification

From equations 2 and 5:

$$(T2 \to T1) = (T5 \to T4)$$

we get:

$$T2 = T5 \qquad \text{and} \qquad T1 = T4$$

Now, from equations 3 and 6:

$$T2 = (Tright + T3)$$

$$T5 = (Ts + Tleft)$$

and since $T2 = T5$, we must unify:

$$(Tright + T3) = (Ts + Tleft)$$

This gives:

$$Tright = Ts \qquad \text{and} \qquad T3 = Tleft$$

From equation 4:

$$T3 = (Ts \to T4)$$

Therefore:

$$Tleft = (Ts \to T4)$$

Now substitute into equation 6:

$$T5 = (Ts + Tleft)$$

So:

$$T5 = (Ts + (Ts \to T4))$$

Since $T2 = T5$, we also get:

$$T2 = (Ts + (Ts \to T4))$$

Now we can compute the type of Tf. From equation 2:

$$Tf = (T2 \to T1)$$

Using

$$T2 = (Ts + (Ts \to T4))$$

and

$$T1 = T4$$

we get:

$$Tf = ((Ts + (Ts \to T4)) \to T4)$$

Finally, from equation 1:

$$T0 = (Tf \to T1)$$

Using

$$Tf = ((Ts + (Ts \to T4)) \to T4)$$

and

$$T1 = T4$$

we get:

$$T0 = (((Ts + (Ts \to T4)) \to T4) \to T4)$$

Final substitution

$$\begin{aligned}
T1 &= T4 \\
T3 &= (Ts \to T4) \\
Tright &= Ts \\
Tleft &= (Ts \to T4) \\
T5 &= (Ts + (Ts \to T4)) \\
T2 &= (Ts + (Ts \to T4)) \\
Tf &= ((Ts + (Ts \to T4)) \to T4)
\end{aligned}$$

Final inferred type

$$T0 = (((Ts + (Ts \to T4)) \to T4) \to T4)$$

Final result

The inference succeeds.

The final type of the expression is:

$$(((Ts + (Ts \to T4)) \to T4) \to T4)$$